Optimization and Curve Sketching

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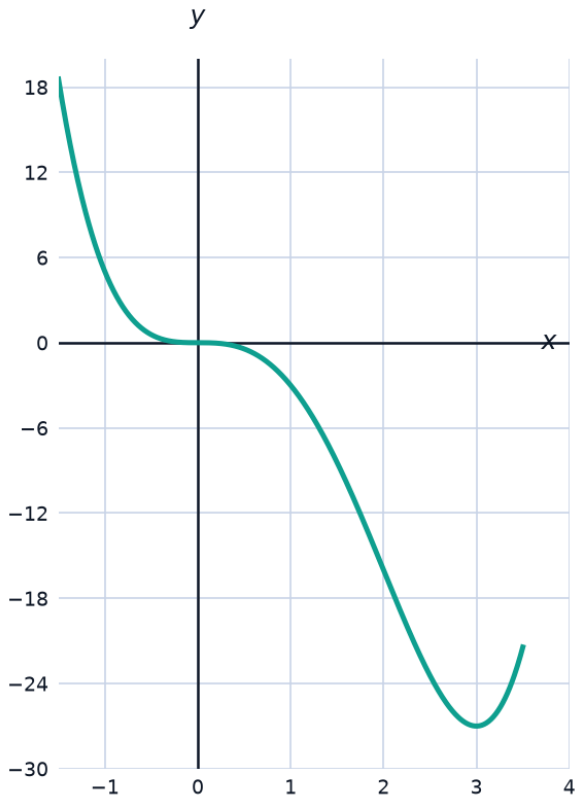
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

2 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

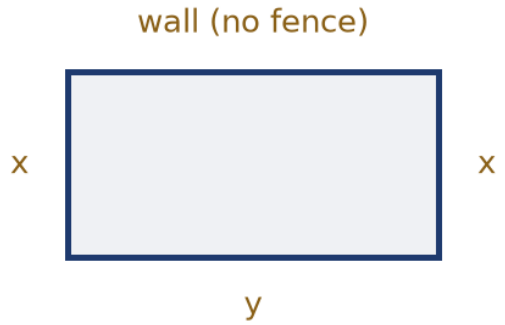
For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.



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3 [Foundational] • Target: Optimization applications

A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

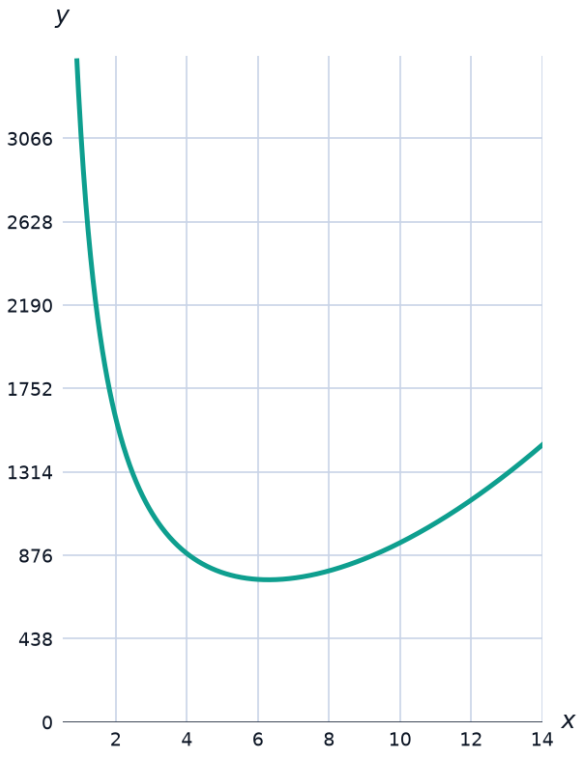


4 [Foundational] • Target: Optimization applications

An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

5 [Foundational] • Target: More optimization

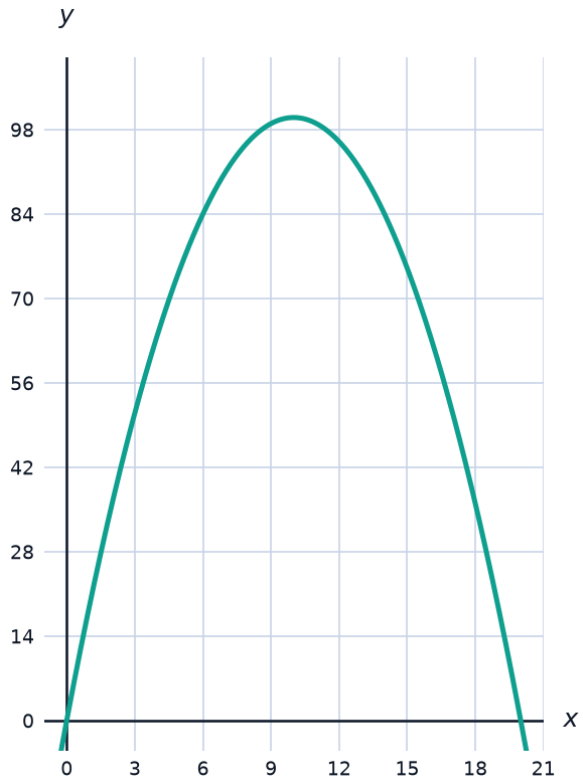
A cylindrical can must have volume 500π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).



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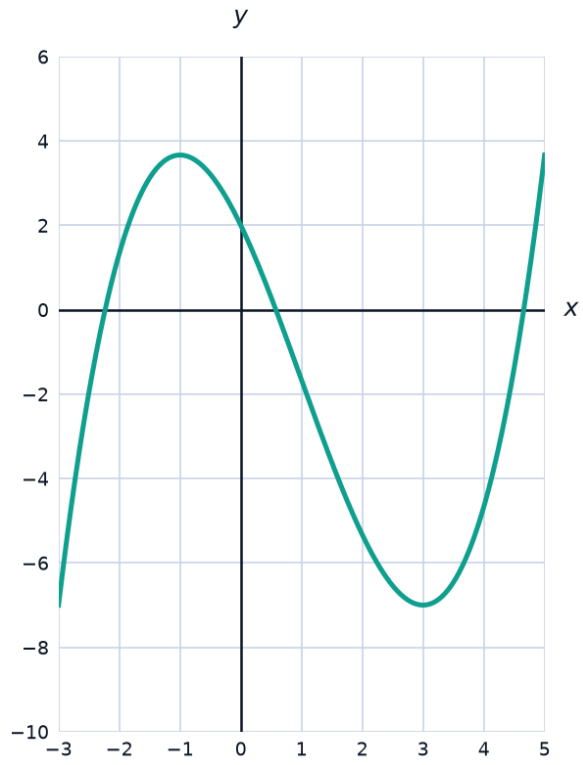
6 [Foundational] • Target: More optimization

Find two positive numbers whose sum is 20 and whose product is as large as possible.



7 [Foundational] • Target: Curve sketching

For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.

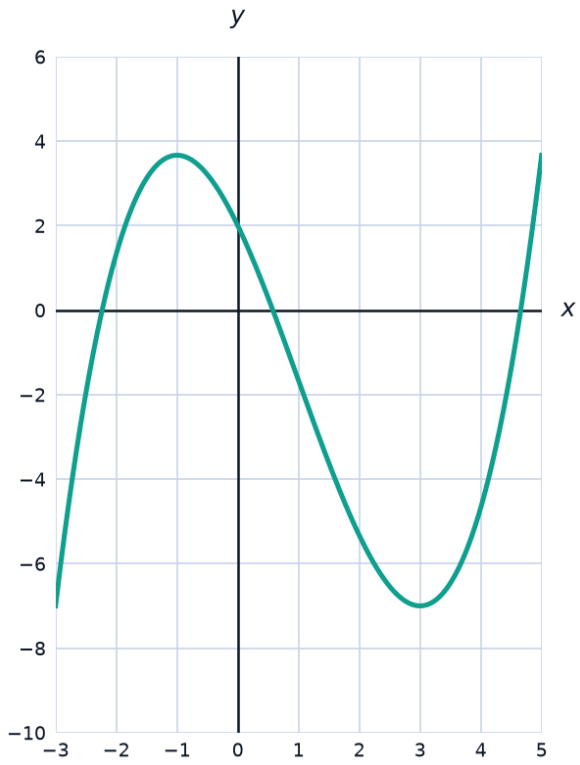


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Curve sketching**

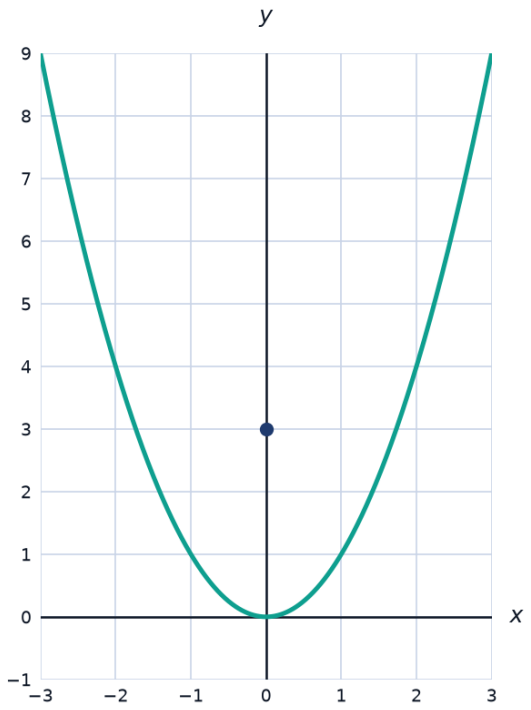
Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.



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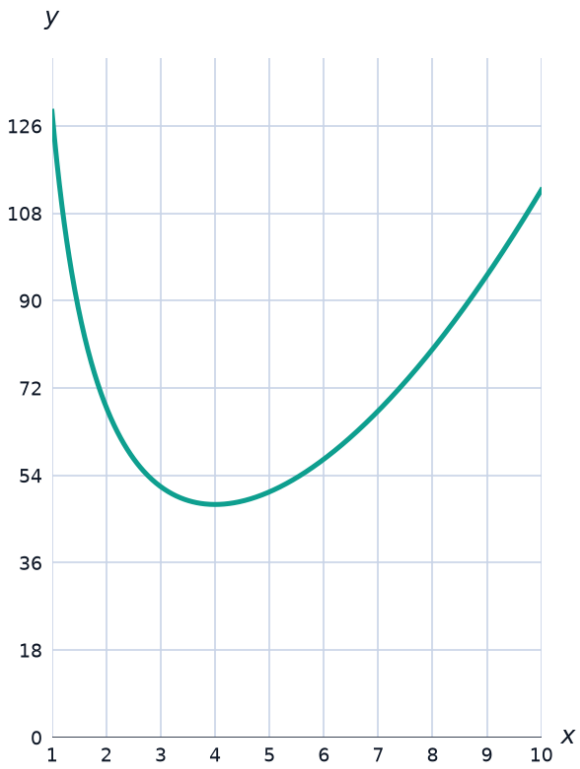
9 [Intermediate] • Target: Minimizing distance and cost

Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.



10 [Intermediate] • Target: Minimizing distance and cost

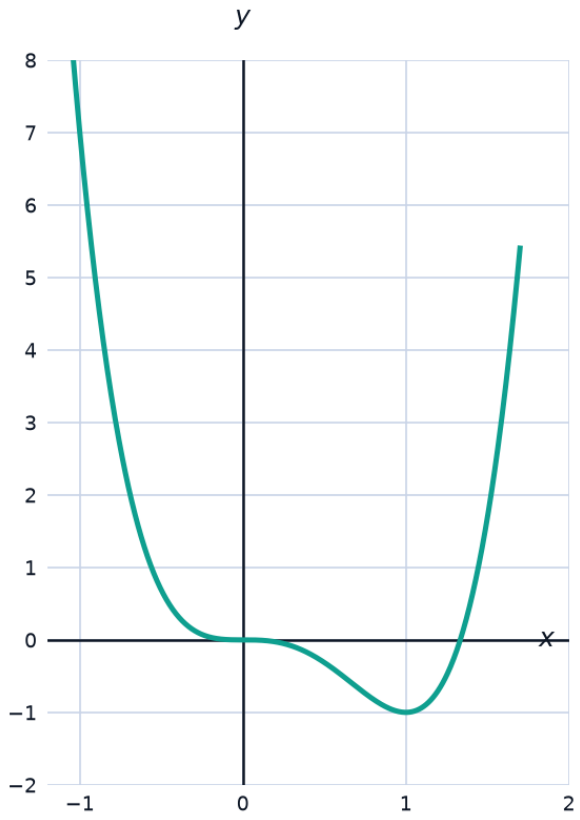
A rectangular box with a square base and open top must have volume 32 m^3. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2h = 32$).



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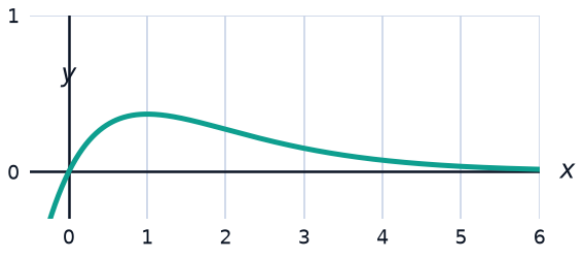
11 [Intermediate] • Target: Sketching using derivative information

For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.



12 [Intermediate] • Target: Sketching using derivative information

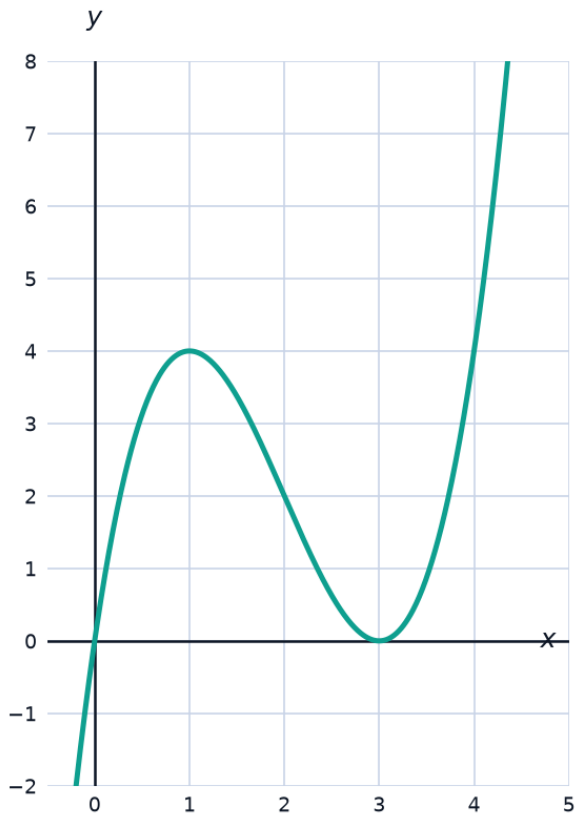
For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.



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13 **[Intermediate]** • **Target: Visualizing critical points on a graph**

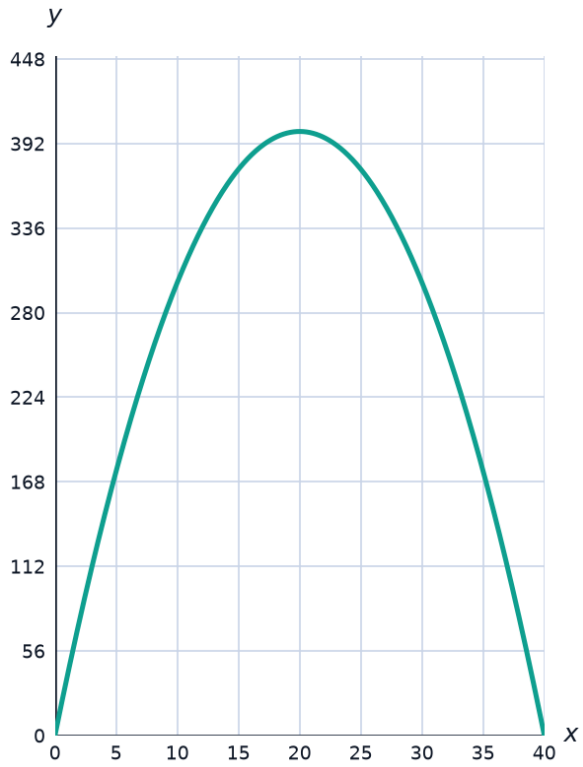
The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.



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14 [Intermediate] • Target: Optimization with a fixed perimeter

A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

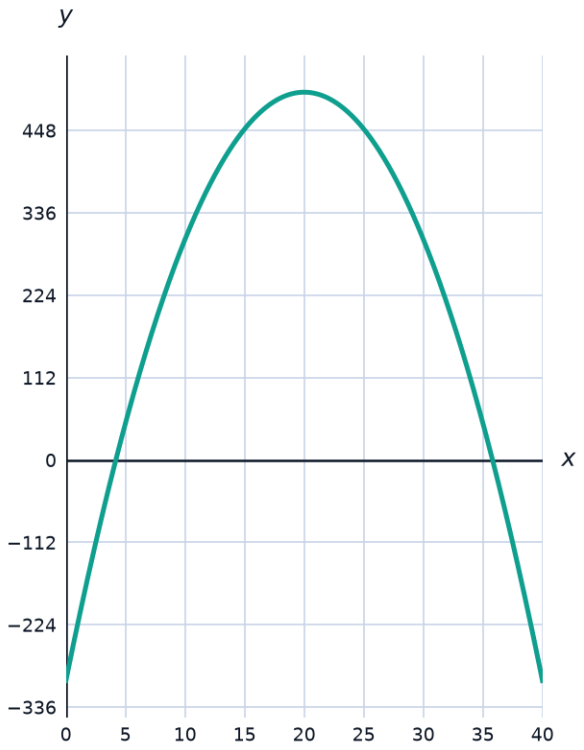
15 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 16.01 Vector Fields and Line Integrals

A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y: $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).

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16 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 2.01 Evaluating Limits

A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.



17 [Challenging] • Target: Curve sketching • Cross-Lesson: 2.02 Continuity

Local min at $x = c$ by First Derivative Test requires f':.

18 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 3.01 The Derivative from First Principles

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:.

19 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:.

20 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Critical Points and First Derivative Test

Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. $x = 1$ and $x = 2$

b. $x = -1$ and $x = -2$

c. $x = 3$ and $x = 0$

d. $x = 2$ only

22 [Foundational] • Target: Critical Points and First Derivative Test

Classify the local extrema of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. Local max at $x = 1$; local min at $x = 2$

b. Local min at $x = 1$; local max at $x = 2$

c. Inflection point at $x = 1$

d. Local max at $x = 2$ only

23 [Foundational] • Target: Critical Points and First Derivative Test

Find the absolute maximum of $g(x) = 4x - x^2$ on the closed interval $[0, 3]$.

a. 4

b. 3

c. 0

d. 5

24 [Foundational] • Target: Concavity and the Second Derivative Test

Determine the interval(s) where $f(x) = x^4 - 4x^3 + 10$ is concave upward.

a. $(-\infty, 0) \cup (2, \infty)$

b. $(0, 2)$

c. $(2, \infty)$

d. $(-\infty, 2)$

25 [Foundational] • Target: Concavity and the Second Derivative Test

Find the x-coordinates of all inflection points of $y = x^3 - 6x^2 + 9x$.

a. $x = 2$

b. $x = 1$ and $x = 3$

c. $x = 0$

d. $x = 4$

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26 [Foundational] • Target: Concavity and the Second Derivative Test

If $f'(3) = 0$ and $f''(3) = -5$, what does the Second Derivative Test conclude about $x = 3$?

a. f has a local maximum at $x = 3$

b. f has a local minimum at $x = 3$

c. $x = 3$ is an inflection point

d. The test is inconclusive

27 [Foundational] • Target: Applied Optimization: Perimeter, Area, and Volume

A farmer has 120 m of fencing to enclose a rectangular field adjacent to a river (no fence needed along the river). What dimensions maximize the enclosed area?

a. Width $= 30$ m, Length $= 60$ m

b. Width $= 40$ m, Length $= 40$ m

c. Width $= 20$ m, Length $= 80$ m

d. Width $= 25$ m, Length $= 70$ m

28 [Foundational] • Target: Applied Optimization: Perimeter, Area, and Volume

An open-top rectangular box is made from a 16 by 10 inch cardboard sheet by cutting equal squares of side x from each corner and folding up the sides. What x maximizes the box volume?

a. 2 inches

b. 3 inches

c. 1.5 inches

d. 2.5 inches

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Applied Optimization: Perimeter, Area, and Volume

A closed cylindrical can must hold 1000 cm^3 of liquid. What relationship between height h and radius r minimizes total surface area?

a. $h = 2r$

b. $h = r$

c. $h = 4r$

d. $h = \pi r$

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30 [Intermediate] • Target: Economics and Distance Optimization

A company's revenue and cost functions are $R(x) = 50x - 0.5x^2$ and $C(x) = 10x + 200$. What production level x maximizes profit?

a. 40

b. 50

c. 30

d. 60

31 [Intermediate] • Target: Economics and Distance Optimization

Find the point on the line $y = 2x + 1$ that is closest to the point $(4, 4)$.

a. $(2, 5)$

b. $(1, 3)$

c. $(3, 7)$

d. $(1.5, 4)$

32 [Intermediate] • Target: Economics and Distance Optimization

A Norman window has the shape of a rectangle surmounted by a semicircle. If the perimeter is 16 ft, what radius r of the semicircle maximizes the light admitted?

a. $\frac{16}{4+\pi}$ ft

b. $\frac{8}{2+\pi}$ ft

c. 2 ft

d. $\frac{16}{\pi}$ ft

33 [Intermediate] • Target: Economics and Distance Optimization

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. indeterminate without further analysis

c. always min

d. always max

34 [Intermediate] • Target: Critical points and the first/second derivative tests

Optimal rectangle of fixed perimeter P has max area when:

a. length $= 2 \times$ width

b. it is a square

c. width $= P$

d. length $= P/4$

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35 [Intermediate] • Target: Optimization applications

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

a. $\frac{26}{3}$

b. $\frac{28}{3}$

c. $\frac{25}{3}$

d. $\frac{27}{3}$

36 [Intermediate] • Target: More optimization

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx}f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. saddle point

b. local max

c. local min

d. inconclusive

37 [Intermediate] • Target: Curve sketching

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2 + y^2} =$

a. always 1

b. an integer

c. always 0

d. always π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 3.01 The Derivative from First Principles

Second derivative of $f(x) = x^4$:

a. $4x^3$

b. $12x^2$

c. $12x$

d. $24x$

39 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

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40 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$

d. $\pi e^{4\pi}$

41 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. inconclusive

42 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find all critical points of $f(x) = x^{2/3}(x - 2)$:

a. $x = 0$ only

b. $x = 2$ only

c. $x = 0$ and $x = 4/5$

d. $x = 0$ and $x = 2$

43 [Challenging] • Target: Critical Points and First Derivative Test • Cross-Lesson: 6.01 Related Rates

Fence three sides of rectangle next to wall (wall is 4th side). Area $= xy = A$, perimeter $= x + 2y = L$. Max area:

a. $L^2/8$

b. $L^2/4$

c. $L^2/6$

d. $L^2/2$

44 [Challenging] • Target: Concavity and the Second Derivative Test • Cross-Lesson: 7.01 Integration by Substitution

Critical points occur where $f'(x)$ is:

a. positive

b. negative

c. zero or undefined

d. equal to $f(x)$

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45 [Challenging] • Target: Applied Optimization: Perimeter, Area, and Volume • Cross-Lesson: 7.02 Integration by Parts

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. local min

b. local max

c. inflection

d. saddle

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